

ADVIS: An End-to-End Traffic Accident Analysis System Using Unsupervised Anomaly Detection

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Abstract—Traffic accident analysis is a complex problem that requires not only detecting abnormal events but also interpreting their context and extracting meaningful evidence. This study presents ADVIS (Accident Detection and Vehicle Identification System), an end-to-end intelligent traffic analysis platform designed to operate on surveillance video streams.

The proposed system adopts a triggered pipeline architecture in which lightweight perception and anomaly detection run continuously, while computationally expensive diagnostic modules are activated only when suspicious behavior is detected. The anomaly detection component is based on an unsupervised LSTM autoencoder trained exclusively on normal traffic data, enabling the system to identify abnormal motion patterns without requiring labeled accident datasets.

Following anomaly detection, the system performs higher-level analysis using a multimodal large language model to generate structured accident interpretations, including event type and contextual reasoning. This analysis is further enriched with license plate recognition, allowing the extraction of identity-level evidence from video data.

In addition to its analytical capabilities, ADVIS is implemented as a full-stack platform with a secure backend and a real-time frontend dashboard, supporting incident management, monitoring, and user interaction.

Experimental results demonstrate that the proposed unsupervised anomaly detection approach can effectively identify abnormal traffic behavior and serve as a reliable trigger for downstream analysis. Overall, the system illustrates how traffic accident analysis can be addressed as an integrated pipeline combining perception, reasoning, and software infrastructure.

I. INTRODUCTION

A. Background and Motivation

Traffic accidents represent a major global issue, leading to significant loss of life, property damage, and economic impact. With the increasing number of vehicles and the complexity of urban traffic environments, monitoring and analyzing traffic incidents has become a critical challenge. Traditionally, accident detection and analysis rely heavily on manual observation, human reporting, and post-incident investigation, which are often time-consuming, error-prone, and inconsistent.

Recent advancements in computer vision and artificial intelligence (AI) have enabled the development of intelligent systems capable of analyzing video streams in real time. In particular, anomaly detection techniques and deep learning-based models have shown promising results in identifying unusual patterns in dynamic environments such as traffic scenes. However, many existing systems focus solely on detecting anomalies without providing deeper contextual understanding, such as identifying the type of accident, or extracting forensic evidence.

To address these limitations, there is a growing need for integrated systems that not only detect traffic incidents automatically but also perform detailed analysis and generate structured, actionable reports suitable for real-world and legal applications.

B. Problem Definition

Despite the availability of surveillance cameras and traffic monitoring systems, current solutions often lack the ability to autonomously detect, interpret, and document traffic accidents in a comprehensive manner. Existing approaches typically suffer from one or more of the following limitations:

- Inability to detect accidents in real time with high accuracy
- Lack of contextual understanding of the detected events
- Absence of automated legal interpretation
- Limited integration with backend systems for data management and reporting
- Inadequate support for extracting critical information such as vehicle identity and license plate data

Therefore, the main problem addressed in this project is the design and implementation of an end-to-end intelligent system that can automatically detect traffic anomalies, analyze accident scenarios, extract license plate information, and generate structured forensic reports.

C. Objectives of the Study

The primary objective of this project is to develop a comprehensive and modular system, named ADVIS (Accident-Vehicle Identification System), that performs automated traffic accident detection and analysis using video data.

The specific objectives of the study are as follows:

- To design a real-time anomaly detection pipeline for identifying abnormal traffic behavior
- To develop an unsupervised anomaly detection model capable of identifying abnormal traffic behavior using only normal traffic data
- To implement a post-anomaly analysis module capable of interpreting accident scenarios
- To integrate license plate recognition (LPR) and optical character recognition (OCR) techniques for vehicle identification
- To build a secure and scalable backend system for incident management and data storage
- To design a user-friendly frontend dashboard for real-time monitoring and interaction
- To generate structured, forensic-quality reports suitable for legal and operational use

D. Contributions of the Project

This project makes several key contributions to the field of intelligent traffic monitoring systems:

- End-to-End System Design: Development of a complete pipeline from video input to forensic report generation
- Unsupervised Anomaly Detection Approach: Implementation of an LSTM autoencoder for detecting abnormal traffic behavior based on learned normal patterns
- Hybrid AI Integration: Combination of classical computer vision techniques with large language models (LLMs) for structured accident interpretation
- Motion-Based Anomaly Scoring: Use of tracking-derived motion features to identify potential crash events in video data
- Modular and Scalable Architecture: Design of a system that integrates AI, backend, and frontend components in a cohesive manner
- Real-Time Communication: Implementation of WebSocket-based live notifications for incident monitoring
- Forensic Evidence Handling: Use of hashing, timestamps, and structured outputs to ensure data integrity and traceability

E. Organization of the Paper

The remainder of this paper is organized as follows:

- Section II reviews existing literature on traffic accident detection, anomaly detection, and related technologies.
- Section III presents the overall system architecture and describes the interaction between system components.
- Section IV details the anomaly detection module based on an unsupervised LSTM autoencoder approach.

- Section V explains the accident analysis and interpretation module based on large language models.
- Section VI describes the license plate recognition system and OCR pipeline.
- Section VII discusses the backend system architecture and implementation details.
- Section VIII describes the frontend dashboard and user interface design.
- Section IX presents experimental results and performance evaluation.
- Section X concludes test results.
- Section XI provides a discussion of the system's strengths and limitations.
- Section XII concludes the paper and outlines potential future work.

II. RELATED WORK

A. Traffic Accident Detection Systems

Traffic accident detection has been an active area of research in intelligent transportation systems. Traditional approaches relied on sensor-based systems, including loop detectors, radar, and GPS data, to identify abnormal traffic conditions. While these systems can provide useful signals, they are often limited by infrastructure requirements and lack visual context.

With the widespread deployment of surveillance cameras, vision-based accident detection systems have gained significant attention. Early methods focused on rule-based approaches, such as detecting sudden stops, abnormal trajectories, or vehicle overlaps. However, these approaches are highly sensitive to predefined thresholds and struggle to generalize across different traffic scenarios.

More recent works utilize deep learning techniques to detect accidents directly from video streams. Convolutional Neural Networks (CNNs) and 3D CNN architectures have been applied to capture spatiotemporal features from video data [1]. While these methods improve detection performance, they often require large labeled datasets and are computationally expensive, making them less suitable for real-time edge deployment.

B. Video Anomaly Detection Approaches

Video anomaly detection aims to identify events that deviate from normal patterns without explicitly defining all possible abnormal scenarios. This problem is particularly relevant in traffic environments, where unexpected events such as collisions, illegal maneuvers, or sudden stops may occur.

Unsupervised and semi-supervised approaches are widely used in this domain. One of the most common methods is based on autoencoders, where a model is trained to reconstruct normal data. During inference, abnormal events produce higher reconstruction errors, which are used as anomaly scores [2]. Recurrent architectures such as Long Short-Term Memory (LSTM) networks are frequently employed to capture temporal dependencies in sequential data.

Another line of work focuses on predictive models that forecast future frames or motion patterns. Deviations between

predicted and actual observations indicate anomalies [3]. Although effective, these approaches often operate on raw pixel data, making them sensitive to noise and computationally intensive.

Recent studies have also explored feature-based anomaly detection, where motion trajectories, object interactions, and scene-level dynamics are used instead of raw images. This approach reduces computational complexity and improves interpretability, particularly in structured environments like traffic scenes.

C. Object Detection and Tracking in Traffic Analysis

Object detection and multi-object tracking are fundamental components of modern traffic analysis systems. State-of-the-art object detectors such as YOLO (You Only Look Once) provide real-time performance and high accuracy for detecting vehicles in video frames [4].

For tracking, algorithms such as SORT [5], DeepSORT [6], and ByteTrack [7] are widely used to maintain object identities across frames. ByteTrack, in particular, has demonstrated strong performance by effectively handling low-confidence detections and maintaining track continuity.

Trajectory-based analysis builds upon tracking results to model vehicle behavior over time. Features such as velocity, acceleration, direction, and interaction with neighboring vehicles can be derived from trajectories and used for higher-level reasoning tasks, including anomaly detection and behavior classification.

D. License Plate Recognition Systems

License Plate Recognition (LPR) systems are widely used in traffic monitoring, law enforcement, and toll collection. A typical LPR pipeline consists of two main stages: license plate detection and character recognition.

Recent approaches employ deep learning-based object detectors, such as YOLO, to localize license plates within images. Once the plate region is extracted, Optical Character Recognition (OCR) techniques are applied to decode the alphanumeric content. Modern OCR systems, including EasyOCR and CRNN-based models, leverage deep neural networks for improved robustness [8].

Post-processing techniques, such as regex-based validation and character correction, are often used to improve accuracy, especially in challenging conditions like low lighting, motion blur, and occlusion.

E. AI-Based Accident Analysis and Forensic Systems

Beyond detection, understanding the context of a traffic accident is crucial for decision-making and legal processes. Recent advancements in large language models (LLMs) have enabled new possibilities for interpreting visual data and generating structured descriptions [9].

Multimodal AI systems can analyze video content and produce detailed textual summaries, including event descriptions and inferred causes. These systems can be guided using prompt engineering techniques to generate outputs in

structured formats such as JSON, making them suitable for downstream processing.

However, most existing approaches focus on general video understanding and lack domain-specific reasoning capabilities required for traffic analysis, such as legal interpretation.

F. Limitations of Existing Approaches

Although significant progress has been made in traffic monitoring and video analysis, existing systems still exhibit several limitations:

- Many anomaly detection methods rely solely on visual features and lack interpretability
- Supervised models require large annotated datasets, which are difficult to obtain for rare events like accidents
- Few systems provide end-to-end solutions that combine detection, interpretation, and reporting
- License plate recognition systems often struggle under real-world conditions
- Limited integration between AI models and backend systems for data management and real-time monitoring
- Lack of support for generating structured, forensic-quality outputs suitable for legal applications

To address these challenges, this project proposes a comprehensive system that integrates anomaly detection, accident interpretation, license plate recognition, and full-stack system design into a unified framework.

III. SYSTEM ARCHITECTURE

A. Overview of the ADVIS System

The Accident Detection and Vehicle Identification System (ADVIS) is designed as an end-to-end intelligent traffic surveillance and forensic analysis platform. Its primary purpose is to automatically detect abnormal traffic events, identify potential accidents, analyze the circumstances of the incident, extract vehicle-related evidence, and present the results through a secure and interactive software infrastructure.

Unlike conventional traffic monitoring systems that rely on either manual supervision or isolated computer vision models, ADVIS integrates multiple layers of functionality into a unified architecture. These layers include real-time perception, anomaly detection, post-event accident analysis, license plate recognition, backend evidence management, and frontend visualization. In this way, the system does not merely detect unusual motion patterns, but also transforms them into interpretable and actionable incident reports.

From an architectural perspective, ADVIS is designed as a modular and scalable system. Each major subsystem is responsible for a clearly defined stage of the workflow, while communication between subsystems enables the formation of a continuous processing pipeline.

B. Triggered Pipeline Architecture

A fundamental design principle of ADVIS is the adoption of a triggered pipeline architecture, where continuous monitoring tasks are separated from computationally expensive forensic analysis tasks.

During the continuous monitoring stage, the system performs lightweight but high-frequency operations such as object detection, vehicle tracking, and motion-based anomaly scoring. These tasks run continuously on incoming traffic video streams and provide a low-latency representation of the current traffic scene.

When the anomaly score exceeds a predefined threshold, the system transitions into the diagnostic stage. At this point, more advanced analysis modules are activated, including accident interpretation, license plate recognition, and structured report generation. This design ensures that computationally expensive operations are executed only when necessary.

C. Main Architectural Components

The ADVIS architecture is composed of three principal layers: the AI and Computer Vision Layer, the Backend Management Layer, and the Frontend Presentation Layer.

1) *AI and Computer Vision Layer*: The AI and Computer Vision Layer constitutes the analytical core of the system. Its responsibility is to process raw video data and transform it into structured information about vehicle behavior and traffic incidents.

The first stage of this layer is object detection and multi-object tracking. Vehicles are detected using YOLO-based object detection models, while their identities are maintained across frames using ByteTrack. This produces temporally consistent trajectories for each observed vehicle.

The second stage is anomaly detection. ADVIS employs an unsupervised LSTM autoencoder trained exclusively on normal traffic sequences. The model learns to reconstruct normal motion patterns, and anomalies are identified based on reconstruction error. This approach enables the system to detect abnormal traffic behavior without requiring labeled accident data.

Once an anomalous segment is identified, the system activates post-event analysis modules. These include accident interpretation using a multimodal large language model and license plate detection with OCR. The output of this layer is a structured interpretation of the incident, including accident type, involved vehicles, and contextual information.

2) *Backend Management Layer*: The Backend Management Layer is responsible for converting AI-generated outputs into a secure and manageable service infrastructure. It is implemented using a Spring Boot-based architecture.

This layer handles authentication, authorization, incident persistence, camera monitoring, notification delivery, and evidence integrity. JWT-based authentication and role-based access control mechanisms ensure secure system access. The backend also supports a multi-tenant architecture, enabling logical separation between different organizations.

Incident data, including timestamps, identifiers, and evidence references, are stored in a PostgreSQL database. Features such as UUID-based identification and hashing mechanisms ensure forensic traceability and data integrity.

Real-time communication is supported via WebSocket/STOMP, allowing immediate notification of newly detected incidents and system events.

3) *Frontend Presentation Layer*: The Frontend Presentation Layer provides the user interface for interacting with the ADVIS platform. It is implemented using Next.js, React, and TypeScript.

The frontend includes modules for live monitoring, incident review, camera management, and system status visualization. Role-based access ensures that users are presented only with relevant information based on their permissions.

Real-time updates from the backend are handled via WebSocket connections, allowing dynamic interface updates without page reloads. The interface follows a modular dashboard structure, emphasizing usability, clarity, and responsiveness.

D. End-to-End Data Flow

The overall system operates as a multi-stage pipeline from video input to structured output.

Video streams are ingested and processed through object detection and tracking. Motion features derived from these trajectories are evaluated by the anomaly detection module.

If no anomaly is detected, the system continues monitoring the scene. When an anomaly threshold is exceeded, the corresponding video segment is forwarded to the analysis stage.

In this stage, accident interpretation and license plate recognition are performed. The results are then stored in the backend and transmitted to the frontend for real-time visualization and user interaction.

E. Deployment Considerations

ADVIS is designed with real-world deployment constraints in mind. Computationally intensive tasks such as object detection, tracking, and sequence modeling are executed on GPU-enabled environments, while backend services operate on CPU-based infrastructure.

Efficient data exchange between system components is achieved through asynchronous processing and low-latency communication protocols. Temporary storage mechanisms are used for buffering video streams and preserving event-related evidence.

F. Architectural Summary

The ADVIS architecture transforms continuous traffic monitoring into an event-driven analysis system. By combining real-time perception, unsupervised anomaly detection, and post-event forensic analysis within a modular framework, the system provides a scalable and practical solution for intelligent traffic incident analysis.

IV. ANOMALY DETECTION MODULE

A. Overview

The anomaly detection module constitutes the first analytical stage of ADVIS and is responsible for identifying abnormal traffic events from continuous video streams. As the system operates in real time, this module functions as the triggering

mechanism that determines when computationally expensive post-event analysis modules should be activated.

Instead of relying on raw video frames, ADVIS adopts a tracking-based temporal analysis strategy. Vehicles are first detected and tracked across frames, and their motion patterns are transformed into structured feature representations. These representations are then analyzed using sequence models to distinguish normal traffic behavior from abnormal events.

Given the practical constraint that labeled accident data is scarce or unavailable in real-world traffic camera systems, this project adopts an unsupervised anomaly detection approach. The model is trained exclusively on normal traffic data and identifies anomalies as deviations from learned motion patterns.

B. Data Preprocessing

1) *Object Detection and Tracking*: The first step in the anomaly detection pipeline is vehicle detection and tracking. YOLO-based object detectors are used to localize vehicles in each frame, while ByteTrack is employed to maintain consistent identities across frames.

The tracking output includes bounding box coordinates, object identities, and temporal consistency information. From these trajectories, kinematic features such as displacement, velocity, acceleration, and direction are derived.

2) *Motion Feature Extraction*: The tracking output is converted into structured motion features for each tracked vehicle. These features include smoothed center coordinates, displacement, velocity, acceleration, speed magnitude, heading, heading change, bounding box dimensions, and local interaction features such as proximity to neighboring vehicles.

This feature representation captures both individual vehicle dynamics and interactions within the traffic scene, enabling more interpretable anomaly detection compared to raw pixel-based approaches.

3) *Feature Normalization*: To ensure numerical stability, motion features are normalized using z-score standardization after applying quantile-based clipping. The normalization statistics are computed from normal training data and reused during inference.

4) *Sliding Window Generation*: Since temporal modeling is required, motion features are segmented into fixed-length sequences using a sliding window approach. Each window represents a short temporal segment of a vehicle trajectory.

C. Unsupervised LSTM Autoencoder

1) *Model Architecture*: The anomaly detection model is based on an LSTM autoencoder. The encoder processes the input sequence and compresses it into a latent representation. The decoder reconstructs the original sequence from this representation.

The model is trained to minimize reconstruction error, enabling it to learn the underlying distribution of normal traffic behavior.

2) *Training Strategy*: Only normal traffic sequences are used during training. The model is optimized using mean squared error (MSE) loss and the AdamW optimizer. The best model is selected based on validation reconstruction loss.

This training strategy reflects real-world constraints where normal traffic data is abundant but labeled accident data is limited.

3) *Anomaly Scoring*: During inference, the reconstruction error is computed for each input sequence. Higher reconstruction errors indicate a greater likelihood of abnormal behavior.

Window-level anomaly scores are aggregated into frame-level and segment-level scores using statistical aggregation methods such as mean and maximum pooling. These scores are then used to detect and localize anomalous events.

D. Advantages and Limitations

The primary advantage of the unsupervised approach is that it does not require labeled accident data, making it suitable for real-world deployment scenarios. Additionally, the use of structured motion features improves interpretability and reduces computational complexity.

However, the model is sensitive to threshold selection and may produce false positives for unusual but non-dangerous traffic patterns. Furthermore, the performance depends on the quality of tracking and feature extraction.

E. Summary

The anomaly detection module of ADVIS is based on a tracking-centered temporal analysis framework and a single unsupervised LSTM autoencoder model. This design enables scalable and practical detection of abnormal traffic events in environments where labeled accident data is not available.

V. ACCIDENT ANALYSIS AND INTERPRETATION MODULE

A. Overview

Once the anomaly detection module identifies a suspicious temporal segment, ADVIS transitions from low-level motion analysis to high-level semantic interpretation. In this stage, the objective is to explain what happened, identify the involved vehicles, and extract meaningful forensic evidence.

This module operates as part of the triggered pipeline and is activated only after anomaly verification, ensuring computational efficiency during normal operation.

The accident analysis and interpretation module is designed as a hybrid reasoning layer that combines multimodal large language model analysis with auxiliary evidence from OCR. It produces structured outputs that are later integrated into human-readable and machine-readable reports.

B. Video-Based Accident Analysis

The core of this module is implemented through a multimodal large language model pipeline. The analysis begins by processing the event video and submitting it to the model together with a constrained prompt.

Unlike generic video captioning systems, the model is not asked to generate free-form descriptions. Instead, the task is

formulated as a structured reasoning problem. The model is instructed to infer accident type, involved vehicles, environmental conditions, and possible causes.

This design transforms the module into a domain-specific accident interpretation engine rather than a general-purpose summarization system.

C. Large Language Model Integration

The large language model is integrated as a multimodal reasoning component that operates on video input and structured prompts. OCR outputs are processed separately and later fused with the interpretation results.

This separation ensures modularity: the language model focuses on high-level reasoning, while OCR provides low-level identity extraction. Such a design improves robustness and prevents role overlap between different subsystems.

D. Prompt Engineering and Structured Output Design

The system prompt frames the model as a traffic accident analysis expert and constrains the reasoning process using domain-specific guidelines.

A key restriction is that the model is not allowed to read or generate license plates, as this responsibility belongs to the OCR subsystem. This prevents hallucination and ensures consistency between modules.

The output is required in a structured JSON format, including fields such as accident type, accident subtype, summary, environmental conditions, involved vehicles, legal reasoning, and confidence score.

The model is also instructed to leave uncertain fields as null rather than generating unsupported information. This is particularly important in forensic applications.

After inference, the JSON output is parsed and converted into structured internal representations for downstream processing.

E. Accident Type Classification

The model assigns each event to a predefined accident category, such as rear-end collision, side impact, intersection accident, or lane-change violation.

In addition to classification, the system generates a chronological explanation of the event, combining categorical and descriptive outputs to improve interpretability.

F. Legal Interpretation

The interpretation incorporates domain-specific legal reasoning, such as traffic law principles and fault distribution logic. However, the output is intended as decision support rather than a legally binding conclusion.

G. Integration with OCR

OCR is applied to sampled video frames to extract candidate license plates. These outputs are later associated with vehicle descriptions using heuristic matching based on visual attributes such as vehicle type and color.

This integration allows the system to combine semantic interpretation with identity-level evidence.

H. Report Generation

The final analysis is converted into two formats: a human-readable textual report and a structured JSON output.

The report includes accident type, summary, environmental context, involved vehicles and legal interpretation. The JSON format preserves the same information for backend storage and API usage.

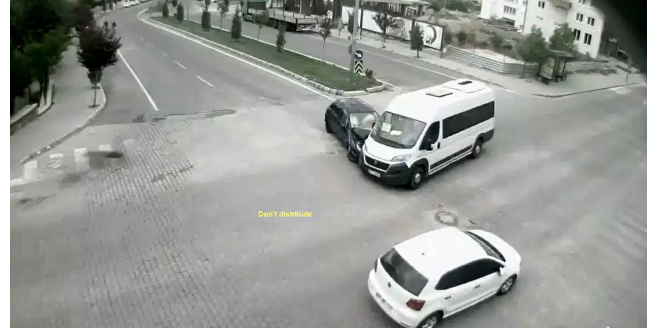


Fig. 1. Sample accident frame detected by the ADVIS system.

For the detected event shown in Fig. 1, the system generated the following interpretation:

“The black vehicle entered the intersection without slowing down and collided with an oncoming minibus, indicating an uncontrolled intersection entry.”

I. Strengths and Limitations

The primary strength of this module is its ability to transform short video segments into structured and interpretable accident reports. By combining multimodal reasoning with structured output design, the system provides more informative results than simple classification approaches.

However, the module has limitations. The reasoning is probabilistic rather than authoritative, OCR-to-vehicle matching is heuristic, and visual ambiguity can affect performance. Therefore, the system should be considered a decision-support tool rather than a replacement for human experts.

J. Summary

The accident analysis and interpretation module extends ADVIS beyond anomaly detection by introducing a semantic reasoning stage. By combining multimodal analysis with structured reporting, it serves as a bridge between low-level detection and high-level forensic interpretation.

VI. LICENSE PLATE RECOGNITION MODULE

A. Overview

The License Plate Recognition (LPR) module is a key forensic component of ADVIS. Its role is to extract the alphanumeric identity of vehicles associated with detected incidents. Within the system architecture, this module is part of the triggered diagnostic pipeline and is activated only after anomaly detection.

Rather than operating as a standalone vision task, LPR is integrated into the broader accident analysis workflow. After an event segment is identified, the system samples frames, detects license plate regions, performs OCR, validates the results according to Turkish plate formats, and integrates the output with the accident interpretation module.

B. Design Objectives

The LPR subsystem is designed with three primary objectives: robustness under real-world conditions, efficiency within the triggered pipeline, and graceful degradation in case of partial failure.

The system is expected to operate under varying lighting conditions, motion blur, and partial occlusions. Additionally, failure in plate recognition should not prevent the overall report generation process.

C. Pipeline Structure

The LPR pipeline consists of the following stages:

- Event frame sampling
- Plate region detection
- OCR-based text recognition
- Post-processing and Turkish plate validation
- Multi-frame aggregation
- Fusion with accident interpretation output

D. Frame Sampling

Instead of processing all frames, the system extracts a fixed number of evenly spaced frames from the event video. This reduces computational cost while maintaining sufficient coverage.

The number of sampled frames is configurable, allowing a trade-off between accuracy and processing speed.

E. Plate Region Detection

Plate detection is performed using a YOLOv8-based model trained for license plate localization. Detected regions are cropped with slight padding to improve OCR accuracy.

As a fallback mechanism, the system can use a classical Haar Cascade detector if the YOLO-based model fails.

F. OCR-Based Recognition

The cropped plate regions are processed using EasyOCR. The OCR module is optimized for efficiency by reusing a singleton model instance and enabling GPU acceleration when available.

Character recognition is constrained using an allowlist tailored to Turkish license plates, improving both speed and accuracy.

If plate crops fail to yield valid results, the system applies OCR to the entire frame as a fallback strategy.

G. Post-Processing and Validation

OCR outputs are refined using domain-specific rules. Recognized text is cleaned and validated against Turkish license plate patterns.

Additional character correction rules are applied to handle common OCR errors such as letter-digit confusion. This step significantly improves recognition reliability.

H. Multi-Frame Aggregation

Since plate visibility may vary across frames, recognition is performed on multiple sampled frames.

Detected plate candidates are aggregated, and the highest-confidence result for each unique plate is retained. This improves robustness and reduces duplicate detections.

I. Result Fusion

Recognized plates are associated with vehicles identified by the accident interpretation module. This is performed using heuristic matching based on attributes such as vehicle type and color.

Although practical, this approach is not fully geometry-based and may introduce ambiguity in complex scenes.

J. Fallback Mechanism

If no valid plate is detected, the system continues report generation without plate information.

K. Strengths and Limitations

The LPR module is robust, modular, and optimized for real-world conditions. It combines modern detection methods with domain-specific validation and multi-frame aggregation.

However, performance may degrade under poor visibility conditions, and plate-to-vehicle association remains heuristic.

L. Summary

The LPR module transforms visual input into structured vehicle identity information. By integrating detection, OCR, validation, and aggregation within a unified pipeline, it supports the generation of evidence-oriented traffic incident reports.

VII. BACKEND SYSTEM ARCHITECTURE

A. Overview

The backend layer of ADVIS serves as the operational and security backbone of the system. While the AI modules are responsible for detecting and interpreting traffic incidents, the backend transforms these outputs into a secure, manageable, and production-ready service infrastructure.

Its primary responsibilities include authentication, authorization, incident management, camera management, evidence persistence, real-time notification delivery, and system monitoring. By bridging analytical outputs with user-facing services, the backend ensures that AI-generated results are accessible, traceable, and operationally meaningful.

The system is implemented using Spring Boot and follows modern enterprise backend design principles, including layered architecture, stateless authentication, and asynchronous processing.

B. Technology Stack

The backend is built using a Java-based enterprise stack. The core framework is Spring Boot, with Java 21 as the implementation language. PostgreSQL is used as the relational database system, while JPA/Hibernate provides object-relational mapping.

Authentication is implemented using JWT-based stateless security. Real-time communication is supported via WebSocket and STOMP. Additional components include ModelMapper for DTO transformations, Jakarta Validation for input validation, and Spring Mail for email services.

This stack balances enterprise-level reliability with the responsiveness required for real-time traffic monitoring systems.

C. Layered Architecture

The backend follows a layered architecture:

- Controllers: Handle HTTP requests and expose REST endpoints
- Services: Contain business logic
- Repositories: Provide data access using JPA
- Entities: Represent persistent data models

Supporting modules such as configuration, DTOs, exception handling, and utility components further improve modularity and maintainability.

D. Authentication and Authorization

Security is implemented using JWT-based stateless authentication. Upon login, the system generates access and refresh tokens containing user-related claims.

Role-Based Access Control (RBAC) is used to enforce authorization policies. Supported roles include ADMIN, OPERATOR, AUDITOR, COMPANY_MANAGER, and LAW_ENFORCEMENT.

Password security is ensured using BCrypt hashing, and token invalidation is supported through blacklist mechanisms.

E. Multi-Tenant Design

The backend adopts a multi-tenant architecture based on organizational separation. Each company acts as a tenant boundary, and related entities such as users, cameras, and incidents are scoped accordingly.

This design enables a single deployment to serve multiple organizations while maintaining strict data isolation.

F. Incident and Evidence Management

The backend manages incident records generated by the AI pipeline. Each incident includes metadata such as timestamps, identifiers, and references to associated evidence.

Forensic traceability is supported through UUID-based identification and hashing mechanisms. Retention policies and scheduled cleanup processes are implemented to manage storage efficiently.

G. REST API Design

System functionality is exposed through a versioned REST API using the `/api/v1` prefix.

The API includes endpoints for authentication, incident management, camera management, user operations, and system configuration. Pagination, validation, and structured error handling are incorporated to ensure robustness and usability.

H. Real-Time Communication

Real-time system updates are delivered using WebSocket and STOMP protocols. The backend can push updates such as new incidents, status changes, and camera activity directly to connected clients.

This design eliminates the need for polling and enables responsive user interfaces.

I. Asynchronous Processing

Non-critical tasks are handled asynchronously using Spring Async. These include email delivery and background maintenance operations.

Scheduled jobs are used for periodic cleanup and retention management, ensuring that the system remains responsive during real-time operation.

J. System Monitoring

The backend includes system monitoring capabilities that provide insights into resource usage, such as CPU and memory.

These metrics support operational awareness and help identify performance bottlenecks in real-time deployments.

K. Strengths and Limitations

The backend architecture is modular, secure, and aligned with real-time system requirements. It supports multi-user access, real-time communication, and forensic data management.

However, large-scale deployment may require further optimization, including distributed system design and enhanced tenant isolation mechanisms.

L. Summary

The backend layer enables ADVIS to function as a complete operational platform by transforming AI-generated outputs into a secure and scalable service infrastructure. It provides the necessary foundation for real-time monitoring, evidence management, and user interaction.

VIII. FRONTEND SYSTEM ARCHITECTURE

A. Overview

The frontend layer of ADVIS provides the primary interaction interface between users and the intelligent traffic analysis platform. While the backend manages data, security, and communication, the frontend is responsible for presenting real-time information, visualizing incidents, and enabling user interaction through a unified dashboard.

Rather than serving as a passive visualization layer, the frontend functions as an active control and monitoring environment, supporting multiple user roles and operational workflows.

B. Technology Stack

The frontend is implemented using Next.js, React, and TypeScript. This combination enables a structured application architecture, component-based UI development, and improved maintainability through static typing.

Styling is handled using Tailwind CSS together with reusable UI components. Additional libraries include tools for form validation, data visualization, icon rendering, and WebSocket communication.

This stack supports a modular, responsive, and real-time dashboard-oriented user experience.

C. Global Application Layout

A centralized layout component defines the global structure of the application. It wraps all pages with shared providers responsible for authentication, real-time communication, and notifications.

Key global components include:

- Authentication provider for session management
- WebSocket provider for live updates
- Notification system for alerts

This design ensures that cross-cutting concerns are handled at a single level, improving consistency and reducing duplication.

D. Authentication and Session Management

Session management is handled through a client-side authentication context that stores access and refresh tokens.

The frontend is responsible for restoring session state, managing token refresh, and exposing user identity and role information to the interface. Unauthorized users are redirected to the login page.

This mechanism complements the backend's JWT-based authentication model.

E. Dashboard-Oriented Interface

The main user interface is designed as a dashboard consisting of:

- A navigation sidebar
- A central content area
- Contextual panels for alerts and system status

This layout supports efficient navigation between system functions such as live monitoring, incident review, and system management.

F. Role-Based Navigation

The interface adapts dynamically based on user roles. Navigation options are filtered so that users only see features relevant to their permissions.

This reduces complexity, improves usability, and reinforces backend authorization policies at the presentation level.

G. Dynamic Content Rendering

The main content area renders different modules depending on the selected section. Each functional component, such as incident views or system monitoring, is implemented as an independent module.

This modular approach improves maintainability and allows independent evolution of system features.

H. Real-Time Communication

The frontend integrates WebSocket-based communication to receive live updates from the backend.

Through this mechanism, the system can:

- Display new incidents in real time
- Update system and camera status dynamically
- Synchronize monitoring views with backend events

This design eliminates the need for polling and ensures responsiveness in operational environments.

I. Reusable UI Components

The interface is built using reusable UI components that standardize layout, interaction patterns, and styling.

This approach ensures consistency across the application and simplifies development of new features.

J. Strengths and Limitations

The frontend architecture is modular, role-aware, and designed for real-time interaction. It supports multiple user roles and provides a consistent dashboard experience.

However, as the system scales, additional considerations such as performance optimization, improved routing strategies, and enhanced state management may be required.

K. Summary

The frontend completes the ADVIS platform by providing a unified interface for monitoring, interaction, and visualization. Through its modular design and real-time capabilities, it enables effective user engagement with system outputs.

IX. EXPERIMENTS AND RESULTS

A. Experimental Setup

The experimental evaluation of ADVIS focuses on validating the effectiveness of the proposed unsupervised anomaly detection pipeline under realistic traffic conditions.

The anomaly detection model was trained using several hours of normal traffic footage captured from a fixed surveillance camera. This setup reflects real-world constraints, where abnormal events such as traffic accidents are rare and labeled data is limited.

For testing, a separate video sequence of approximately 7 minutes from the same camera was used. This test video contains three distinct anomaly segments corresponding to abnormal traffic behavior.

All videos were processed through a unified pipeline consisting of object detection, multi-object tracking, and motion feature extraction. The resulting temporal sequences were used as input to the LSTM autoencoder model.

B. Anomaly Scoring Methodology

The model computes anomaly scores based on reconstruction error. Each temporal segment is assigned a segment-level anomaly score, which reflects the degree of deviation from learned normal behavior.

To enable event detection, a threshold-based decision mechanism is used. Segments exceeding the predefined threshold are classified as anomalous and trigger further analysis in the pipeline.

C. Temporal Anomaly Score Analysis

To analyze the behavior of the model over time, segment-level anomaly scores were plotted across the duration of the test video.

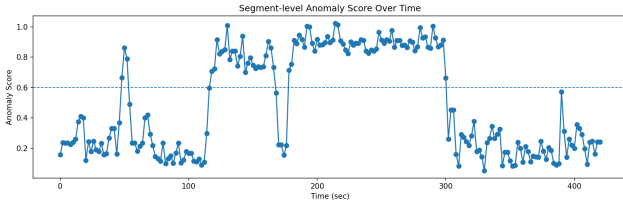


Fig. 2. Segment-level anomaly scores over time. The dashed line indicates the anomaly threshold.

As shown in Fig. 2, the anomaly score remains consistently low during normal traffic conditions. In contrast, three distinct peaks are clearly visible, corresponding to the anomaly segments present in the test video.

These peaks demonstrate that the model successfully detects deviations from normal traffic behavior and localizes anomaly events in time.

D. Top Anomalous Segments

To further examine the model output, the highest-scoring segments were extracted and are summarized in Table I.

TABLE I
TOP ANOMALY SEGMENTS DETECTED BY THE MODEL

Start (sec)	End (sec)	Score	Anomaly
214.0	215.97	1.023	Yes
216.0	217.97	1.013	Yes
130.0	131.97	1.008	Yes
290.0	291.97	1.004	Yes
192.0	193.97	1.004	Yes

The detected segments align closely with the visually observed anomaly regions in the test video. This confirms that high anomaly scores correspond to meaningful abnormal events.

E. Qualitative Results

To complement quantitative analysis, qualitative evaluation was performed using visual inspection of selected frames.

As shown in fig 3. in normal traffic conditions, vehicle motion remains stable and anomaly scores are low. In contrast, anomaly segments show vehicles with unusual behavior, such

as abrupt motion changes or abnormal positioning, which are successfully highlighted by the model.

These qualitative results confirm that the model captures meaningful behavior-level deviations rather than random fluctuations.

F. Discussion of Results

The experimental results demonstrate that the unsupervised anomaly detection model is effective in identifying abnormal traffic events without requiring labeled accident data.

The temporal score analysis shows clear separation between normal and anomalous segments, while the top anomaly segments align with visually confirmed abnormal events. The qualitative results further validate that the model responds to semantically meaningful deviations in vehicle behavior.

However, the model is sensitive to threshold selection and may occasionally respond to rare but non-critical traffic patterns. Additionally, performance depends on the quality of tracking and motion feature extraction.

G. Summary

Overall, the experiments confirm that the proposed unsupervised anomaly detection framework provides a reliable mechanism for detecting abnormal traffic behavior. By combining temporal modeling with reconstruction-based scoring, the system can effectively trigger downstream accident analysis modules in real-world scenarios.

X. TEST RESULTS

This section presents the results of the testing process based on the test plan defined for the ADVIS system. Each test case is briefly described along with its execution result. The outcomes are categorized as **Passed** or **Partially Passed**.

A total of **38 test cases** were defined across all system modules. Table III summarizes the distribution of results.

Passed (29 tests): The majority of the system modules passed their respective tests successfully. All authentication and authorization tests (TC-34 through TC-38) confirmed the security layer operates correctly. Vehicle detection (TC-01, TC-02), multi-object tracking (TC-03), trajectory extraction (TC-05), and anomaly event triggering (TC-09, TC-10) all performed as expected. Collision anomaly detection (TC-07) successfully identified anomalous events and triggered alerts. The full evidence and storage pipeline passed entirely, including evidence generation (TC-18), database storage (TC-19), and the end-to-end pipeline test (TC-20). All six frontend tests (TC-21 through TC-26) passed, confirming that role-based access control, live perception rendering, real-time alert feed, incident archives, and forensic views operate correctly. Backend infrastructure tests including incident trigger processing (TC-27, TC-28), camera heartbeat (TC-29), RTSP ingestion (TC-30), circular buffer integrity (TC-32), and video evidence generation (TC-33) all passed.

Partially Passed (9 tests): Several modules exhibited correct behavior in most scenarios but showed limitations under specific conditions. Tracking consistency (TC-04) was affected

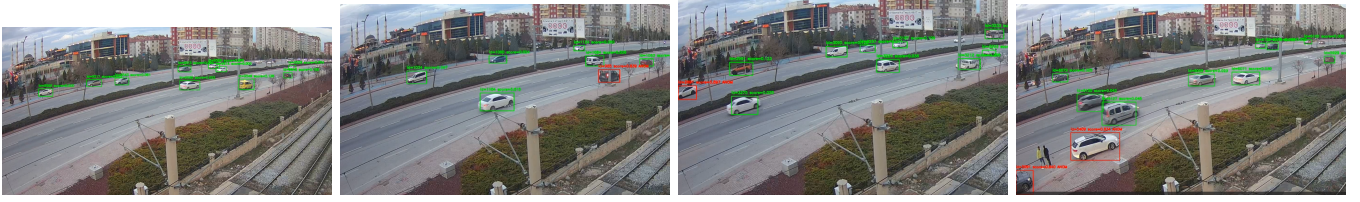


Fig. 3. Comparison between normal traffic (left) and detected anomaly cases (right). Anomalous vehicles are highlighted.

by temporary occlusions in dense traffic scenes. Sudden stop anomaly detection (TC-08) occasionally produced elevated scores for non-anomalous sharp braking maneuvers. Accident classification for both rear-end (TC-11) and side-impact (TC-12) collisions produced correct results in typical cases but showed reduced accuracy in ambiguous scenarios. Faulty vehicle identification (TC-13) correctly identified the responsible vehicle in most cases but requires further refinement for complex multi-vehicle interactions. License plate recognition (TC-15) performed well under favorable conditions but accuracy decreased with low resolution or partial occlusion. Environmental tests for low visibility (TC-16) and high traffic density (TC-17) demonstrated acceptable performance but highlighted the need for additional model tuning. RTSP stream reconnection (TC-31) functioned correctly in most interruption scenarios but requires further hardening for prolonged outages.

Overall Assessment: No test case resulted in a complete failure, and no tests remain in progress, indicating that all system modules have reached a testable state. The 76.3% full pass rate demonstrates the maturity and reliability of the core detection, tracking, frontend, backend, and authentication subsystems. The 9 partially passed tests highlight specific areas where model fine-tuning, threshold calibration, and environmental robustness can be further improved. Future work will focus on improving anomaly detection sensitivity through threshold optimization, enhancing license plate recognition accuracy under adverse visual conditions, and strengthening RTSP reconnection resilience.

XI. DISCUSSION

A. General Discussion

The ADVIS system was designed as a complete intelligent traffic analysis platform rather than a standalone anomaly detection model. Its primary contribution lies in integrating multiple technical components into a unified end-to-end pipeline that spans perception, anomaly detection, semantic interpretation, license plate recognition, and full-stack software infrastructure.

Unlike many existing approaches that stop at anomaly detection, ADVIS extends the processing chain toward structured, interpretable outputs. Detected anomalies are not only flagged but also analyzed, contextualized, and transformed into human-readable and machine-readable reports. This system-level integration represents a key outcome of the project.

B. Strengths of the System

One of the major strengths of ADVIS is its modular architecture. Each subsystem, including tracking, anomaly detection, accident interpretation, license plate recognition, backend services, and frontend visualization, is designed as an independent yet interoperable component. This modularity improves maintainability and allows incremental system evolution.

Another important strength is the triggered processing strategy. The system performs continuous lightweight monitoring and activates computationally expensive modules only when an anomaly is detected. This design improves efficiency and makes the system suitable for real-time deployment scenarios.

The project also demonstrates a hybrid AI approach. It combines tracking-based motion analysis, temporal sequence modeling, OCR, heuristic matching, and large language model reasoning. This division of responsibilities allows each component to address a specific subproblem effectively.

Finally, ADVIS stands out as a full-stack system rather than a model-only prototype. The backend introduces authentication, authorization, multi-tenant organization, and real-time communication, while the frontend provides an interactive dashboard for monitoring and management. This significantly increases the system's practical relevance.

C. Limitations of the Current System

Despite its strengths, the system has several important limitations.

The most critical limitation is the dependency on object detection and tracking quality. Since all downstream modules rely on consistent trajectories, tracking errors can propagate through the pipeline and affect anomaly detection and interpretation accuracy.

Another limitation is scene dependency. The system performance may vary depending on camera placement, visibility conditions, and environmental factors such as lighting, weather, and occlusion. These factors can influence detection, tracking, and OCR reliability.

The anomaly detection model itself is based on an unsupervised learning paradigm. While this makes the system applicable in data-scarce environments, it also introduces sensitivity to threshold selection and reduces specificity. The model may respond to unusual but non-critical traffic patterns, leading to potential false positives.

The accident interpretation module, based on large language models, also has inherent limitations. Its outputs are proba-

TABLE II
SUMMARY OF TEST CASE RESULTS FOR THE ADVIS SYSTEM.

ID	Description	Expected Output	Result
Vehicle Detection & Tracking			
TC-01	Vehicle detection from traffic video frames	Vehicles detected with correct bounding boxes	Passed
TC-02	Detection test on empty road frame	No vehicle detected	Passed
TC-03	Multi-object tracking with multiple vehicles	Each vehicle assigned a unique ID and tracked	Passed
TC-04	Tracking consistency across continuous frames	Vehicle IDs remain consistent across frames	Partially Passed
TC-05	Trajectory extraction from tracking data	Correct trajectory coordinates generated	Passed
Anomaly Detection			
TC-06	Normal traffic behavior (no anomaly expected)	No anomaly detected	Passed
TC-07	Collision anomaly detection	System detects anomaly and triggers alert	Passed
TC-08	Sudden stop anomaly detection	Anomaly score generated correctly	Partially Passed
TC-09	Event trigger upon anomaly detection	Diagnostic pipeline activated	Passed
TC-10	Video clip extraction after trigger	Correct incident video clip extracted	Passed
Accident Analysis & LPR			
TC-11	Rear-end collision classification	Classified as rear-end collision	Partially Passed
TC-12	Side-impact collision classification	Classified as side impact	Partially Passed
TC-13	Faulty vehicle identification	Responsible vehicle correctly identified	Partially Passed
TC-14	License plate detection from vehicle image	License plate detected correctly	Passed
TC-15	License plate character recognition	Correct alphanumeric plate extracted	Partially Passed
Environmental & Stress Tests			
TC-16	Low visibility detection (night/rain)	Vehicles detected with acceptable accuracy	Partially Passed
TC-17	High traffic density scenario	Stable tracking and detection maintained	Partially Passed
Evidence & Storage			
TC-18	Evidence generation after analysis	Evidence package generated successfully	Passed
TC-19	Database storage of incident report	Report correctly stored in database	Passed
TC-20	Full end-to-end pipeline test	Complete pipeline executes successfully	Passed
Frontend			
TC-21	Role-based access control	Access granted/denied based on role	Passed
TC-22	Live perception view rendering	Bounding boxes and anomaly charts render	Passed
TC-23	Real-time alert feed	Alert appears immediately with correct severity	Passed
TC-24	Incident archives table	Records display properly in table	Passed
TC-25	Forensic overview and identity modal	AI summary and LPR data displayed	Passed
TC-26	Forensic analysis and evidence view	Incident evidence displayed correctly	Passed
Backend & Infrastructure			
TC-27	Backend incident trigger processing	Incident entry created and stored	Passed
TC-28	Backend trigger retry mechanism	Retry succeeds after connection restored	Passed
TC-29	Camera heartbeat processing	Camera status updated correctly	Passed
TC-30	RTSP video ingestion from live stream	Frames captured and stored in buffer	Passed
TC-31	RTSP stream reconnection after interruption	System reconnects automatically	Partially Passed
TC-32	Circular buffer integrity	Buffer maintains correct frame count	Passed
TC-33	Video evidence generation from buffer	Valid clip generated around detected event	Passed
Authentication & Authorization			
TC-34	User registration	New account created and stored	Passed
TC-35	Duplicate email registration prevention	Registration rejected	Passed
TC-36	User login with valid credentials	User authenticated and access granted	Passed
TC-37	Login with invalid credentials	Login rejected, access denied	Passed
TC-38	Unauthorized role access attempt	Action blocked	Passed

TABLE III
TEST RESULT STATISTICS.

Status	Count	Percentage
Passed	29	76.3%
Partially Passed	9	23.7%
Failed	0	0.0%
Total	38	100%

bilistic and should be considered as decision support rather than authoritative conclusions. This is particularly important in forensic contexts.

Similarly, the license plate recognition pipeline relies on heuristic association between detected plates and vehicles.

In complex scenes with multiple similar vehicles, this may introduce ambiguity.

D. Real-World Applicability

Despite these limitations, ADVIS demonstrates strong practical potential.

The system is suitable for traffic monitoring centers, where operators need to supervise multiple camera feeds and quickly identify abnormal events. The triggered pipeline reduces cognitive load by highlighting relevant segments and providing contextual analysis.

It is also applicable in law enforcement and traffic management scenarios, where structured incident summaries and extracted evidence can support decision-making.

In addition, the system may be useful in insurance and forensic workflows as a preliminary analysis tool that accelerates incident review and documentation.

The presence of a backend system with role-based access control and multi-tenant organization further strengthens its applicability in institutional environments.

E. Scalability Considerations

From a system architecture perspective, ADVIS includes several elements that support scalability, such as modular design, stateless authentication, asynchronous processing, and real-time communication.

The triggered pipeline also reduces computational overhead by avoiding continuous execution of expensive analysis modules.

However, large-scale deployment would require further improvements, including performance optimization, distributed system design, and extensive testing under high-load conditions.

F. Broader Significance

A broader contribution of this project is the demonstration of how intelligent traffic analysis can be approached as a multi-layer system rather than a single machine learning task.

ADVIS integrates computer vision, sequence modeling, multimodal reasoning, backend engineering, and frontend design into a cohesive platform. This highlights the importance of system-level thinking in real-world AI applications.

G. Summary

In summary, ADVIS presents a modular and practical approach to intelligent traffic incident analysis. Its strengths lie in system integration, triggered processing, hybrid AI design, and full-stack implementation.

At the same time, limitations such as tracking dependency, environmental sensitivity, and probabilistic reasoning define the boundaries of the current system and point toward future improvements.

XII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study presented ADVIS (Accident Detection and Vehicle Identification System), an end-to-end intelligent traffic analysis platform designed to detect, interpret, and report traffic incidents from surveillance video streams.

Unlike conventional traffic-monitoring approaches that focus solely on detection, ADVIS was developed as a complete system that integrates anomaly detection, accident interpretation, license plate recognition, backend evidence management, and a role-based frontend dashboard within a unified architecture.

A central contribution of the project is the triggered pipeline design. In this architecture, lightweight perception and anomaly detection operate continuously, while more computationally intensive diagnostic modules are activated only when

suspicious behavior is detected. This improves computational efficiency and supports real-time operation.

The anomaly detection component is based on an unsupervised LSTM autoencoder trained on normal traffic sequences. By modeling normal behavior and identifying deviations through reconstruction error, the system provides a practical solution for anomaly detection in environments where labeled accident data is limited.

Beyond anomaly detection, the system introduces a semantic interpretation stage using a multimodal large language model. This stage generates structured accident descriptions, performs fault reasoning, and produces machine-readable outputs. The integration of license plate recognition further enhances the system by providing identity-level evidence.

From a software engineering perspective, ADVIS is implemented as a full-stack platform. The backend provides authentication, authorization, incident management, and real-time communication, while the frontend offers a modular dashboard for monitoring and interaction.

Overall, the project demonstrates that traffic incident analysis can be approached as a complete multi-layer system that combines perception, reasoning, evidence extraction, and software infrastructure.

B. Main Contributions

The main contributions of this work can be summarized as follows:

- **End-to-End Intelligent System:** Development of a complete pipeline from video input to structured incident reports
- **Triggered Processing Architecture:** Efficient activation of computationally expensive modules based on anomaly detection
- **Unsupervised Anomaly Detection:** Use of an LSTM autoencoder trained on normal traffic data
- **Hybrid Analysis Framework:** Integration of computer vision, OCR, and large language model reasoning
- **Full-Stack Implementation:** Deployment of backend and frontend systems for real-world usability

C. Future Work

Although ADVIS demonstrates a functional and comprehensive framework, several directions remain for future improvement.

1) *Improving Detection and Tracking Robustness:* Future work may focus on improving object detection and tracking under challenging conditions such as occlusion, dense traffic, and sudden motion changes. This would directly improve the reliability of downstream modules.

2) *Dataset Expansion:* The system would benefit from larger and more diverse datasets, including variations in lighting, weather, camera angles, and traffic scenarios. More diverse data would improve generalization and robustness.

3) *Improved Plate-to-Vehicle Association:* Future work could strengthen the association between detected license plates and tracked vehicles by incorporating spatial-temporal matching rather than relying on heuristic methods.

4) *Refinement of the Interpretation Module*: The large language model-based reasoning stage could be improved through more constrained generation, better grounding, and potential human-in-the-loop validation for sensitive cases.

5) *Scalability and Deployment*: Further work is required to support large-scale deployment, including performance optimization, distributed system design, and advanced monitoring tools.

6) *Explainability and Human Oversight*: Future systems may incorporate more explainability features, such as visual anomaly explanations and structured confidence reporting, as well as human review mechanisms.

D. Final Remarks

ADVIS demonstrates that intelligent traffic analysis is not merely a machine learning problem, but a multi-disciplinary system design challenge.

By integrating computer vision, temporal modeling, multimodal reasoning, and full-stack software engineering, the project provides a strong foundation for future research and real-world deployment in intelligent traffic monitoring systems.

APPENDIX A CODE AVAILABILITY

The implementation of the ADVIS system is publicly available:

<https://github.com/FineTunersEtu>

The repository includes all core components of the system, including anomaly detection, accident analysis, license plate recognition, backend services, and frontend implementation.

This resource is provided to support reproducibility and further research.

APPENDIX B SUPPLEMENTARY VISUALIZATION

This appendix provides screenshots of the ADVIS frontend dashboard, illustrating the system's main interface components including live monitoring, incident review, and camera management modules.

Fig. 4. Screenshot 1 — [ADVIS login page with username/password authentication and encrypted connection indicator]

Fig. 5. Screenshot 2 — [Company registration request form for onboarding new organizations to the ADVIS platform]

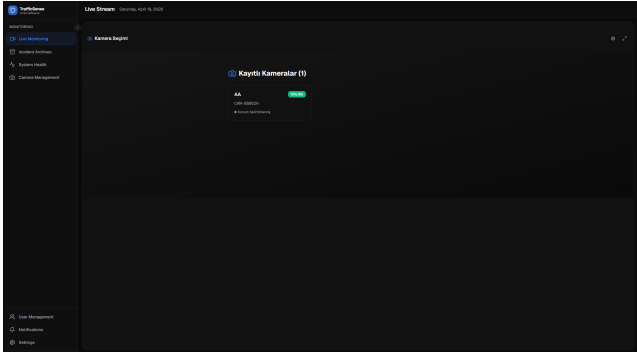


Fig. 6. Screenshot 3 — [Dashboard camera selection view showing registered cameras with online/offline status indicators]

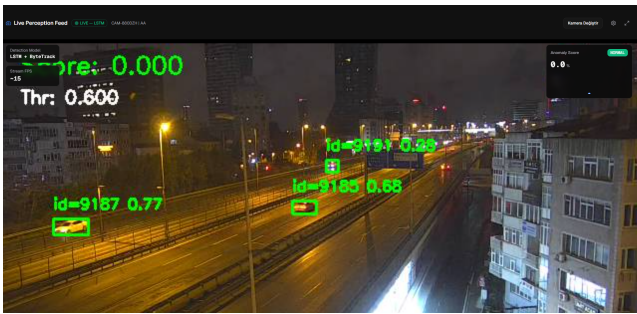


Fig. 7. Screenshot 4 — [Live perception feed with LSTM + ByteTrack vehicle detection, tracking IDs, confidence scores, and real-time anomaly score panel]

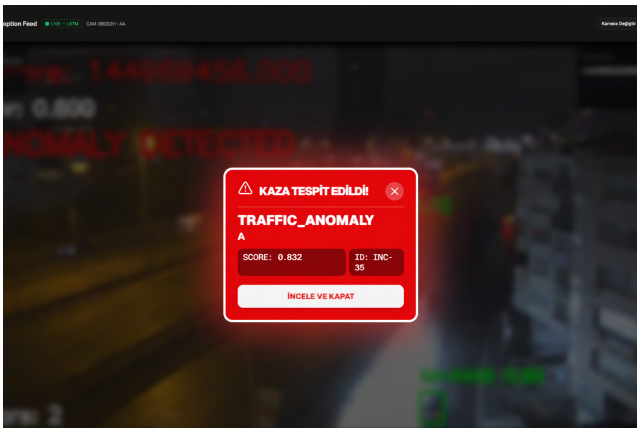


Fig. 8. Screenshot 5 — [Real-time anomaly alert popup triggered by the system, displaying incident type, anomaly score (0.832), and incident ID]

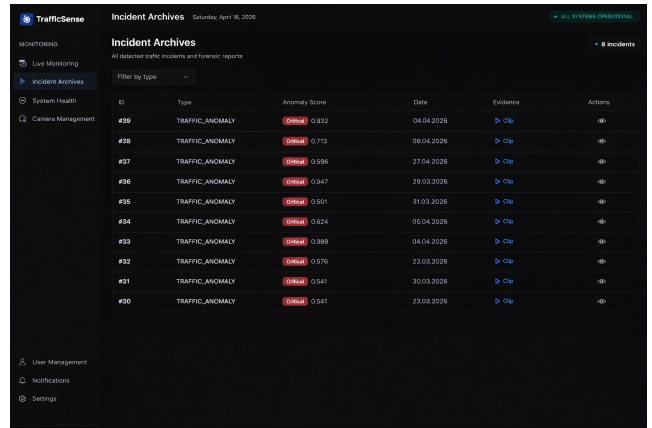


Fig. 9. Screenshot 6 — Incident archives view listing detected traffic anomalies with severity levels, anomaly scores, timestamps, and evidence clip links]

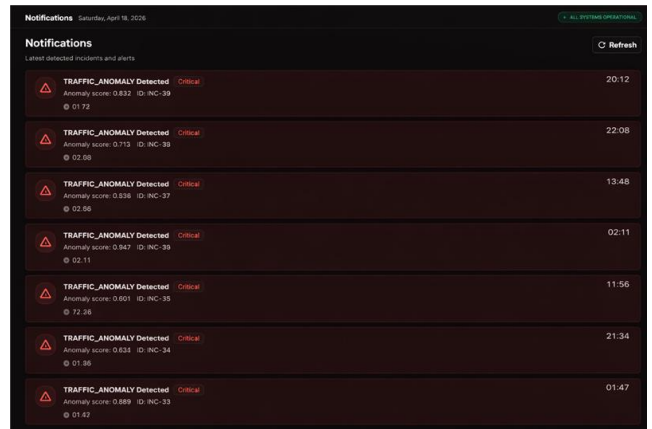


Fig. 10. Screenshot 7 — [Notifications panel showing chronologically ordered incident alerts with anomaly scores, incident IDs, and timestamps]

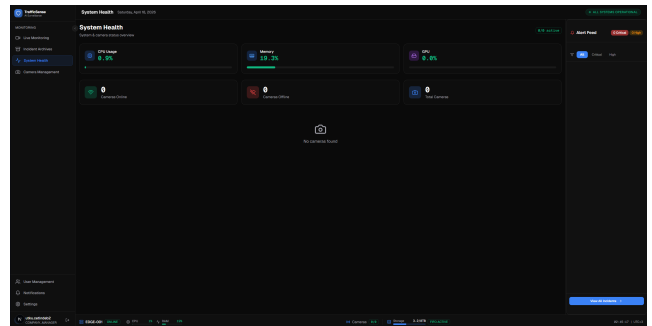


Fig. 11. Screenshot 8 — [System health monitoring dashboard displaying CPU, memory, and GPU usage metrics with camera status summary and edge device information]

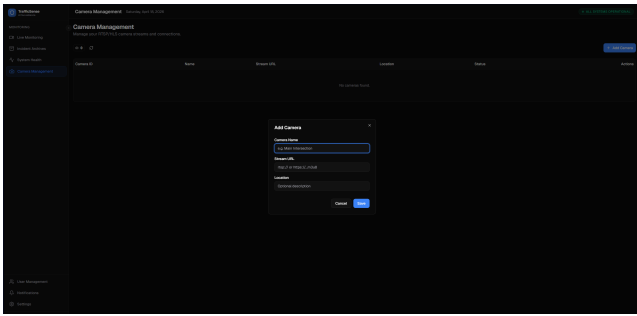


Fig. 12. Screenshot 9 — [Camera management interface with the add camera dialog for registering RTSP/HLS stream sources]

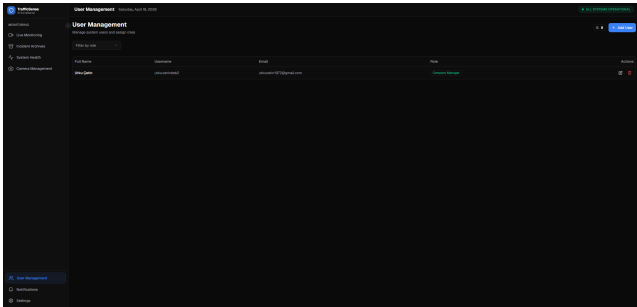


Fig. 13. Screenshot 10 — [User management panel showing registered users with role assignments and administrative actions]

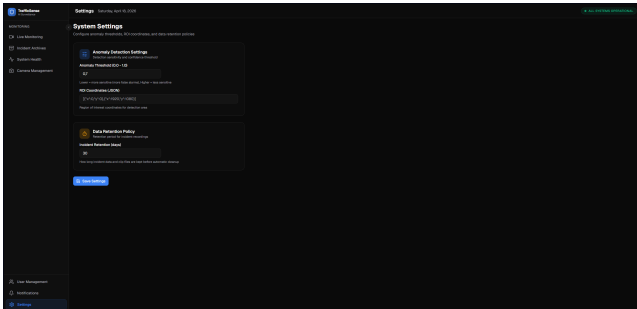


Fig. 14. Screenshot 11 — [System settings page for configuring anomaly detection threshold, ROI coordinates, and data retention policy]

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